📘 Lecture 30- Layer 3 (Multilayer) Switches & SVI Configuration

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🧠 1️⃣ What is a Layer 3 (Multilayer) Switch?

A **Layer 3 Switch** is a special type of switch that can perform **both switching and routing**.

Normally:

- **Layer 2 Switch** → works using **MAC Address**
- **Router / Layer 3 Device** → works using **IP Address**

A **Multilayer Switch** can do **both jobs**.

✅ **Switching (Layer 2)**
✅ **Routing (Layer 3)**

So it is called **Layer 3 Switch** or **Multilayer Switch**.

⚙️ Key Features of Layer 3 Switch

Feature	Explanation
Layer 3 Aware	Understands IP addresses
Switching	Forwards frames using MAC address
Routing	Routes packets using IP address
Supports SVIs	Can create virtual interfaces for VLANs
Inter-VLAN Routing	Allows communication between VLANs

🧩 Why Do We Need Layer 3 Switch?

In normal networks:

- **Switch** → works only inside VLAN
- **Router** → required for communication between VLANs

But a **Layer 3 switch removes the need for a router for VLAN routing.**

It performs **routing internally**.

📦 Real Life Example

Imagine a **large office building**.

Each department is on a different floor:

- Engineering
- HR
- Sales

Each floor is a **VLAN**.

If employees want to talk to another department:

Old Method (Router on a Stick)

PC → Switch → Router → Switch → Destination PC

Traffic travels **multiple times**, wasting resources.

New Method (Layer 3 Switch)

PC → Layer 3 Switch → Destination PC

Routing happens **inside the switch itself**.

⚡ Faster
⚡ Efficient
⚡ Less network load

🔄 Layer 3 Switch = Switching + Routing

Function	How it Works
Switching	Uses MAC Address
Routing	Uses IP Address

When packet arrives:

1️⃣ Switch checks **MAC address for switching**
2️⃣ If packet needs another VLAN → uses **IP address to route**

📊 Router on a Stick (ROAS) vs SVI

Feature	Router on a Stick	SVI (Layer 3 Switch)
Routing Device	Router	Layer 3 Switch
Interface	One trunk interface	Virtual interface for each VLAN
Traffic Path	Switch → Router → Switch	Switch handles routing
Performance	Slower	Faster
Resource Usage	High	Low

Router on a Stick (ROAS)

In **ROAS**:

- Router uses **one physical interface**
- Interface is divided into **multiple subinterfaces**

Example:

G0/0.10
G0/0.20
G0/0.30

Traffic flow:

PC → Switch → Router → Switch → PC

This wastes bandwidth.

SVI (Switch Virtual Interface)

In **Layer 3 Switch**:

Each VLAN gets a **virtual interface**.

Example:

interface vlan 10
interface vlan 20
interface vlan 30

Switch routes traffic **internally**.

🌐 Switch Virtual Interface (SVI)

An **SVI** is a **virtual Layer 3 interface** created for a VLAN.

It allows the switch to assign **IP address to VLAN**.

Example

interface vlan 10
ip address 10.0.0.62 255.255.255.192
no shutdown

Now VLAN 10 has a **gateway IP address**.

PCs inside that VLAN will use it as **default gateway**.

🖥️ Host Gateway Configuration

Hosts must use **SVI IP address** as their **gateway**.

Example:

VLAN	Gateway
VLAN 10	10.0.0.62
VLAN 20	10.0.0.126
VLAN 30	10.0.0.190

Now all VLAN routing is done by **Layer 3 switch**.

🔧 Important Commands

Command	Purpose
ip routing	Enables routing on switch
no switchport	Converts switchport to routed port
interface vlan	Creates SVI
ip route	Creates static route
default interface	Reset interface configuration

🧪 LAB SETUP OVERVIEW

Initially:

Network used:

- **Router R1**
- **Switch SW1**
- **Switch SW2**

But **SW2 was replaced with Layer 3 switch**.

Also **router subinterfaces were removed**.

🧪 Issue with Switch Model

Initially:

Switch **2960** was used.

❌ Problem:

2960 supports **only Layer 2**

So **Layer 3 commands didn't work**.

Solution:

Use **3560-24PS switch**

✔ Supports Layer 3 routing
✔ Supports SVIs

🧪 VLAN Configuration on SW2

```
Switch>en
Switch#conf t
Switch(config)#vlan 10
Switch(config-vlan)#name Engineering
```

Switch(config)#vlan 20
Switch(config-vlan)#name HR

Switch(config)#vlan 30
Switch(config-vlan)#name Sales

🧪 Access Port Configuration

SW2(config)#int fa0/3
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 10

SW2(config)#int fa0/4
switchport mode access
switchport access vlan 10

SW2(config)#int fa0/5
switchport mode access
switchport access vlan 20

These ports connect **PCs to VLANs**.

🧪 Trunk Configuration on SW1

SW1(config)#int fa0/5
SW1(config-if)#switchport mode trunk

Allow specific VLANs:

switchport trunk allowed vlan 10,20,30

Verify trunk:

do show interface trunk

Output shows:

- trunk status
- allowed VLANs
- active VLANs

🧪 Native VLAN Configuration

switchport trunk native vlan 1001

Native VLAN carries **untagged traffic**.

⚠️ Trunk Encapsulation Error

Error message:

Command rejected: An interface whose trunk encapsulation is "Auto"

Reason:

Switch doesn't know whether to use:

- **ISL**
- **802.1Q**

Solution:

Manually define encapsulation.

switchport trunk encapsulation dot1q
switchport mode trunk

🧪 LAB QUESTION 1

Clearing Router Configuration

Old **ROAS configuration must be removed**.

R1(config)#no interface g0/0.10
R1(config)#no interface g0/0.20
R1(config)#no interface g0/0.30

This removes subinterfaces.

Reset interface:

default interface g0/0

Purpose:

Reset interface to **default configuration**.

🧪 LAB QUESTION 2

Configure Layer 3 Port on SW2

Reset interface:

default interface fa0/2

Enable routing:

ip routing

Convert port to routed port:

interface fa0/2
no switchport

Assign IP:

ip address 10.0.0.193 255.255.255.252

🧪 LAB QUESTION 3

Default Route Configuration

Default route sends unknown traffic to router.

ip route 0.0.0.0 0.0.0.0 10.0.0.194

Meaning:

If destination not known → send to **R1**.

Check routing table:

show ip route

Output includes:

Gateway of last resort is 10.0.0.194

This means **default gateway configured successfully**.

🧪 LAB QUESTION 4

SVI Configuration

Create VLAN interface.

VLAN 10

interface vlan 10
ip address 10.0.0.62 255.255.255.192
no shutdown

VLAN 20

interface vlan 20
ip address 10.0.0.126 255.255.255.192
no shutdown

VLAN 30

interface vlan 30
ip address 10.0.0.190 255.255.255.192
no shutdown

These interfaces act as **default gateways**.

🧪 Creating Unknown SVI (VLAN 40)

interface vlan 40
ip address 40.40.40.40 255.255.255.0
no shutdown

But status becomes:

down/down

📊 Conditions for SVI to be UP

For SVI to become **UP/UP**:

✓ VLAN must exist
✓ VLAN must not be shutdown
✓ SVI must not be shutdown
✓ At least one access port in that VLAN must be UP

If these conditions fail → interface remains **down/down**.

🧪 Testing Network Connectivity

Test using:

ping

Example:

PC in VLAN 10 → PC in VLAN 20

If successful:

Layer 3 switch is performing **inter-VLAN routing**.

📊 Important Verification Commands

1️⃣ show ip interface brief

Purpose:

Displays **IP addresses and interface status**.

Example output:

Interface	IP Address	Status	Protocol
Fa0/2	10.0.0.193	up	up
Vlan10	10.0.0.62	up	up

Meaning:

- Interface active
- IP configured

2️⃣ show ip route

Purpose:

Displays **routing table**.

Example:

C 10.0.0.192 is directly connected
S* 0.0.0.0/0 via 10.0.0.194

Meaning:

C → Connected route
S → Static route

- → Default route

3️⃣ show interface status

Displays:

- port status
- VLAN
- speed
- duplex

Example:

Fa0/3 connected vlan 10
Fa0/4 connected vlan 10
Fa0/5 connected vlan 20

Meaning:

Ports assigned to VLANs.

📌 Final Network Operation

After full configuration:

✔ PCs communicate within VLAN
✔ PCs communicate across VLAN
✔ Layer 3 switch performs routing
✔ Router handles external network traffic

🎯 Final Summary

Topic	Description
Layer 3 Switch	Performs switching + routing
SVI	Virtual interface for VLAN
ip routing	Enables routing
no switchport	Converts port to routed port
Default route	Sends unknown traffic to router
Inter-VLAN Routing	Communication between VLANs

✅ VLANs created
✅ Trunks configured
✅ Router configuration cleared
✅ Layer 3 routing enabled
✅ SVIs configured
✅ Default route set
✅ Connectivity tested

Network working successfully 🎉

🧠 1️⃣ Chapter Summary

🔹 What is a Layer 3 (Multilayer) Switch?

A **Layer 3 Switch** is a network device that performs both:

Function	Description
Layer 2 Switching	Forward frames using MAC address
Layer 3 Routing	Route packets using IP address

Normally:

Device	Works Using
Layer 2 Switch	MAC Address
Router	IP Address

A **Layer 3 switch combines both capabilities**.

✔ Switching
✔ Routing

This is why it is called a **Multilayer Switch**.

⚙️ Key Features of Layer 3 Switch

Feature	Description
Layer 3 Awareness	Understands IP addresses
Switching	Uses MAC addresses
Routing	Uses IP routing table
Supports SVIs	Virtual interfaces for VLANs
Inter-VLAN Routing	Communication between VLANs

🧩 Why Layer 3 Switch is Needed

In traditional networks:

PC → Switch → Router → Switch → PC

Router performs inter-VLAN routing.

Problems:

❌ More latency
❌ Bandwidth waste
❌ More hardware usage

With Layer 3 Switch

PC → Layer 3 Switch → PC

Routing occurs **inside the switch**.

Benefits:

⚡ Faster communication
⚡ Less network congestion
⚡ Better performance

🔄 Layer 3 Switch = Switching + Routing

When a packet arrives:

1️⃣ Switch checks MAC address
2️⃣ If destination VLAN is same → switch forwards frame
3️⃣ If destination VLAN is different → switch performs routing using IP address

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📊 Router on a Stick vs Layer 3 Switch

Feature	Router on a Stick	Layer 3 Switch
Device	Router	Multilayer Switch
Interface Type	Subinterfaces	SVIs
Performance	Slower	Faster
Traffic Path	Switch → Router → Switch	Switch routes internally

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🌐 Switch Virtual Interface (SVI)

An **SVI (Switch Virtual Interface)** is a virtual interface created for a VLAN.

It allows the switch to assign an **IP address to a VLAN**.

Example:

```
interface vlan 10
ip address 10.0.0.62 255.255.255.192
no shutdown
```

This IP becomes the **default gateway for hosts in that VLAN**.

🖥 Host Gateway Configuration

Each VLAN uses its **SVI IP as gateway**.

VLAN	Gateway
VLAN 10	10.0.0.62
VLAN 20	10.0.0.126
VLAN 30	10.0.0.190

Hosts must configure this IP as their **default gateway**.

🔧 Important Layer 3 Switch Commands

Command	Purpose
ip routing	Enables routing
interface vlan	Creates SVI
no switchport	Converts port to routed port
ip route	Configures static route
default interface	Reset interface

📊 SVI Status Conditions

For an SVI to be **UP/UP**, these conditions must be met:

✓ VLAN must exist
✓ VLAN must be active
✓ SVI must not be shutdown
✓ At least one access port in VLAN must be active

Otherwise interface status will be:

down/down

🔍 Important Verification Commands

1️⃣ Show Interface Status

show interface status

Displays:

- Port status
- VLAN assignment
- Speed and duplex

2️⃣ Show IP Interfaces

show ip interface brief

Displays:

- Interface IP
- Status
- Protocol state

3️⃣ Show Routing Table

show ip route

Displays routing information.

Example:

C 10.0.0.192 is directly connected
S* 0.0.0.0 via 10.0.0.194

🎯 Chapter Conclusion

Layer 3 switches are **critical devices in modern enterprise networks**.

They provide:

✔ High-speed switching
✔ Fast inter-VLAN routing
✔ Reduced network congestion
✔ Simplified network design

Key ideas:

- Layer 3 switch performs **switching + routing**
- SVIs act as **default gateways for VLANs**
- ip routing enables routing capability
- no switchport converts ports into Layer 3 routed ports
- Default routes send traffic to external networks

Because of their performance and scalability, **Layer 3 switches are widely used in large enterprise networks**.

🎤 Q & A

1️⃣ What is a Layer 3 Switch?

Answer

A **Layer 3 Switch** is a network device that performs both **Layer 2 switching and Layer 3 routing**.

It forwards frames using MAC addresses and routes packets using IP addresses.

Advantages:

✔ Faster than routers for VLAN routing
✔ Reduces network latency
✔ Handles inter-VLAN routing internally

2️⃣ What is a Multilayer Switch?

Answer

A **Multilayer Switch** is another name for a **Layer 3 switch** because it operates at multiple OSI layers.

Functions:

- Layer 2 switching
- Layer 3 routing

It is commonly used in enterprise networks for **inter-VLAN routing**.

3️⃣ What is SVI?

Answer

SVI stands for **Switch Virtual Interface**.

It is a **virtual Layer 3 interface created for a VLAN**.

Example:

```
interface vlan 10
ip address 10.0.0.62 255.255.255.192
```

Purpose:

✔ Provides default gateway for VLAN
✔ Enables inter-VLAN routing

4️⃣ What command enables routing on a Layer 3 switch?

Answer:

```
ip routing
```

Without this command the switch behaves like a **Layer 2 switch**.

5️⃣ What is the difference between Router on a Stick and SVI?

Feature	ROAS	SVI
Device	Router	Layer 3 switch
Interface	Subinterfaces	Virtual interfaces
Performance	Slower	Faster
Routing location	Router	Switch

6️⃣ What does "no switchport" command do?

Answer:

The command **converts a Layer 2 switch port into a Layer 3 routed port**.

Example:

interface fa0/2
no switchport
ip address 10.0.0.193 255.255.255.252

7️⃣ What is inter-VLAN routing?

Answer

Inter-VLAN routing allows **devices in different VLANs to communicate with each other**.

It can be done using:

- Router on a Stick
- Layer 3 switch

8️⃣ Why is a Layer 3 switch faster than Router on a Stick?

Because:

- Routing happens internally
- No trunk traffic to router
- No additional packet travel

Result:

⚡ Lower latency
⚡ Higher performance

9️⃣ What causes an SVI to remain down/down?

Possible reasons:

❌ VLAN not created
❌ VLAN shutdown
❌ No active ports in VLAN
❌ Interface shutdown

🔟 What command verifies routing table?

show ip route

Displays:

- Connected routes
- Static routes
- Default routes

1️⃣1️⃣ Troubleshooting Question

Question

SVI interface is configured but remains down. Why?

Possible reasons:

1️⃣ VLAN not created
2️⃣ No active port in VLAN
3️⃣ SVI shutdown
4️⃣ Trunk not configured

1️⃣2️⃣ Scenario Question

Question

PC in VLAN 10 cannot communicate with VLAN 20.

Possible reasons:

- ip routing not enabled
- SVI not configured
- Gateway misconfigured
- VLAN not allowed on trunk

⭐ One Liner

Explain SVI in one line

Answer:

"An SVI is a virtual Layer 3 interface on a switch used to assign an IP address to a VLAN and perform inter-VLAN routing."

🎯 Final Key Points

✓ Layer 3 Switch = switching + routing
✓ SVIs provide default gateway for VLANs
✓ ip routing enables routing capability
✓ no switchport converts port to routed port
✓ Default route sends unknown traffic to router
✓ Inter-VLAN routing happens inside switch
